

2	(a) energy = $\frac{1}{2}m\omega^2a^2$ and $\omega = 2\pi f$ $= \frac{1}{2} \times 37 \times 10^{-3} \times (2\pi \times 3.5)^2 \times (2.8 \times 10^{-2})^2$ $= 7.0 \times 10^{-3} \text{ J}$ (allow $2\pi \times 3.5$ shown as 7π)	C1 M1 A0	[2]
	Energy = $\frac{1}{2}mv^2$ and $v = r\omega$ Correct substitution Energy = $7.0 \times 10^{-3} \text{ J}$	(C1) (M1) (A0)	
(b)	$E_K = E_P$ $\frac{1}{2}m\omega^2(a^2 - x^2) = \frac{1}{2}m\omega^2x^2$ or E_K or $E_P = 3.5 \text{ mJ}$ $x = a/\sqrt{2} = 2.8/\sqrt{2}$ or $E_K = \frac{1}{2}m\omega^2(a^2 - x^2)$ or $E_P = \frac{1}{2}m\omega^2x^2$ $= 2.0 \text{ cm}$ (E_K or $E_P = 7.0 \text{ mJ}$ scores 0/3)	C1 C1 A1	[3]
	Allow: $k = 17.9$ $E = \frac{1}{2}kx^2$ $x = 2.0 \text{ cm}$	(C1) (C1) (A1)	

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- (c) (i) graph: horizontal line, y-intercept = 7.0 mJ with end-points of line at +2.8 cm and –2.8 cm B1 [1]
- (ii) graph: reasonable curve with maximum at (0,7.0) end-points of line at (–2.8, 0) and (+2.8, 0) B1 [2]
- (iii) graph: inverted version of (ii) with intersections at (–2.0, 3.5) and (+2.0, 3.5) M1 A1 [2]
(Allow marks in (iii), but not in (ii), if graphs K & P are not labelled)
- (d) gravitational potential energy B1 [1]

- 3 (a) sum of potential energy and kinetic energy of atoms/molecules/particles M1